One reported number, a hidden stack

What gets published

"GRPO reaches **0.856** on GSM8K"

label = the algorithm; everything below is unstated

no attribution path: the number is a property of the stack, not of the label

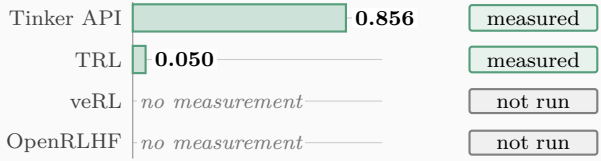
library / framework	*Tinker, TRL, veRL, OpenRLHF*
back-end & hardware	*managed API, H100, CPU*
base checkpoint	*Qwen2.5-0.5B to Qwen3.6-35B*
decoding parameters	*temperature, top-p, max tokens*
group size G	*2, 8, 16, 64*
dataset and split	*gsm8k-500, held-out suites*
seed	*42, 123, 456, 789, 1024*
reward shaping	*verifier, KL coef., length norm.*
advantage normalisation	*group vs. batch*

nine factors, each set per run, all routinely absent from the report

One nominal config, two measured stacks

Qwen3-8B · GRPO · seed 42 · G=8 · lr 10^{-5} · gsm8k-500 · 30 steps

bars: last-10 mean training reward, axis 0–1



The two measured rows sit 17× apart (0.856 vs. 0.050) on a nominally identical configuration. veRL and OpenRLHF were not installed in the sandbox, so no value is shown for them.

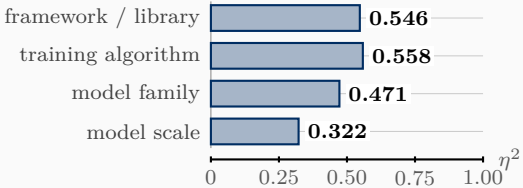
And the label alone does not fix the run

"DAPO" on an open Colab trainer whose dynamic sampling filters zero-variance groups: mean ZVF **0.00**.
"DAPO" on a closed managed stack where only an asymmetric-clip surrogate is settable: mean ZVF **0.58**.
Same label, opposite telemetry.

And the same run gets two answers. The Qwen PPO last-10 GSM8K figure is **22.5%** in the primary results ledger and **35.0%** in the statistical summary of the same campaign.

How much moves with the stack

η^2 over 42 reported experiments (one-way, per factor)



Shares are confounded with one another: do not sum (they total 1.90). They say how much moves with the stack, not why.

Same contrast, opposite answer

PPO vs. GRPO, one run per arm per family; descriptive only, no inferential statistic

Qwen3-8B (provenance conflict): the historical Welch p=0.76 is over the last ten autocorrelated steps of one run per arm; the methods note marks the row *not estimable*, so no effect size is drawn. The arms also ran on different back-ends (GRPO managed, PPO on Modal H100).

Llama-3.1-8B-Instruct: the same table reports the contrast in the opposite direction, again from one run per arm.

The direction is not stable across model families, and neither row supports an algorithm claim.